

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Technical Presentation

Code of Conduct:

*I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A
Technical Presentation

1. Reactive Agent-Based Traffic Control System
Prepare a technical presentation on designing a Reactive Agent for a smart traffic control system. Explain how the agent responds to real-time traffic inputs, manages signal changes, and ensures minimal congestion using streaming data processing.

2. Goal-Based Agent for Healthcare Diagnosis
Develop a presentation on a Goal-Based Agent used in an AI healthcare system. Describe how the agent sets diagnostic goals, evaluates patient data, and selects optimal treatment pathways using decision-making logic and data pipelines.

3. Utility-Based Agent for E-Commerce Recommendations
Create a technical presentation on a Utility-Based Agent that generates personalized product recommendations. Explain how the agent assigns utility values to different options and optimizes user satisfaction using large-scale data processing.

4. Learning Agent for Personalized Education System
Present the design of a Learning Agent in an AI-driven educational platform. Explain how the agent improves performance over time using feedback, adapts content difficulty, and integrates with distributed data processing systems like PySpark.

5. Multi-Agent System for Disaster Management
Prepare a presentation on a Multi-Agent System where multiple agents (sensor agents, decision agents, communication agents) collaborate for disaster detection and response. Explain coordination strategies, data sharing, and real-time decision-making.

1. Reactive Agent-Based Traffic Control System

Slide 1: Title

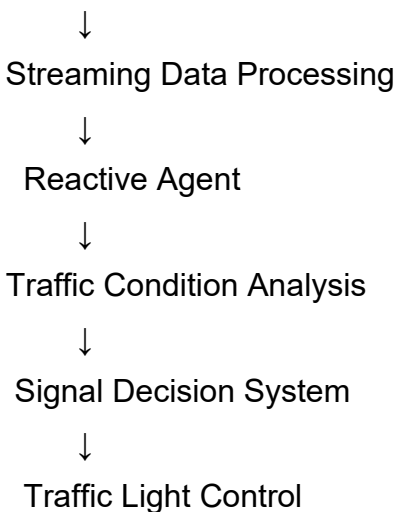
Reactive Agent-Based Smart Traffic Control System

Slide 2: Introduction

- Smart traffic systems use AI to control traffic signals.
- A reactive agent responds immediately to current traffic conditions.
- The system uses real-time data from cameras, sensors, and GPS devices.
- The main objective is to reduce traffic congestion and waiting time.

Slide 3: System Architecture

Traffic Cameras / Sensors / GPS



Slide 4: Working of Reactive Agent

The reactive agent continuously receives real-time traffic information. It checks vehicle density, queue length, and waiting time. Based on predefined rules, it immediately changes traffic signals.

Example:

- High traffic → Increase green signal time.
- Low traffic → Reduce green signal time.
- Emergency vehicle detected → Give priority.

Slide 5: Streaming Data Processing

- Sensors continuously generate traffic data.
- Streaming systems process incoming data in real time.
- Data processing identifies congestion quickly.
- The reactive agent makes immediate decisions.

Slide 6: Benefits

- Reduces traffic congestion.
- Minimizes vehicle waiting time.

- Provides emergency vehicle priority.
 - Improves traffic flow automatically.
-

2. Goal-Based Agent for Healthcare Diagnosis

Slide 1: Title

Goal-Based Agent for AI Healthcare Diagnosis

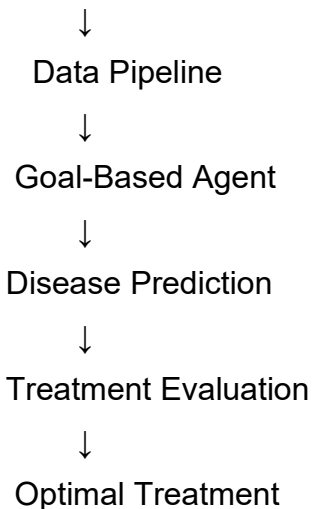
Slide 2: Introduction

A goal-based agent makes decisions by considering a specific objective. In healthcare, the main goal is to identify diseases accurately and select the best treatment.

Slide 3: System Architecture

Patient Data

Symptoms + Reports + Medical History



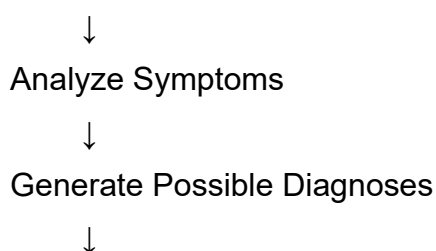
Slide 4: Diagnostic Goals

The agent can have goals such as:

- Identify the possible disease.
- Minimize diagnostic errors.
- Select the safest treatment.
- Improve patient recovery.
- Reduce treatment cost and time.

Slide 5: Decision-Making Process

Collect Patient Data



Evaluate Treatment Options



Select Best Treatment Path



Monitor Patient Response

Slide 6: Advantages

- Supports doctors in decision-making.
- Analyzes large medical datasets.
- Provides faster diagnosis.
- Helps select suitable treatment pathways.

3. Utility-Based Agent for E-Commerce Recommendations

Slide 1: Title

Utility-Based Agent for E-Commerce Recommendations

Slide 2: Introduction

A utility-based agent selects the option that provides the highest benefit or satisfaction. In e-commerce, it recommends products that are most useful and relevant to the user.

Slide 3: System Architecture

User Activity Data



Clicks / Searches / Purchases



Large-Scale Data Processing



Product Analysis



Utility Calculation Agent



Rank Products by Utility



Personalized Recommendations

Slide 4: Utility Calculation

The agent assigns a utility score to each product.

Utility Score =

User Preference

+ Product Rating

- + Purchase History
- + Product Relevance
- Price/Delivery Disadvantages

The product with the highest utility score is recommended first.

Slide 5: Large-Scale Data Processing

Distributed processing can analyze:

- Millions of users.
- Product ratings.
- Search history.
- Purchase transactions.
- Clickstream data.

Technologies such as PySpark can process this data in parallel.

Slide 6: Benefits

- Personalized recommendations.
- Improved user satisfaction.
- Increased product sales.
- Better customer experience.

4. Learning Agent for Personalized Education System

Slide 1: Title

Learning Agent for Personalized Education System

Slide 2: Introduction

A learning agent improves its behavior based on previous experience and feedback. In an educational platform, it learns about each student's strengths, weaknesses, and learning speed.

Slide 3: Learning Agent Architecture

Student Activity



Quiz Scores / Time / Errors



Data Processing System



Learning Agent



Performance Analysis



Adaptive Content Selection



Student Feedback



Continuous Learning

Slide 4: Components of Learning Agent

1. Performance Element – Provides learning recommendations.
2. Learning Element – Improves recommendations using experience.
3. Critic – Evaluates whether recommendations are effective.
4. Problem Generator – Introduces new learning strategies.

Slide 5: Adaptive Learning Process

Student Takes Quiz



Analyze Score



Identify Weak Topic



Reduce Content Difficulty



Provide Learning Materials



Collect Feedback



Update Student Model

If a student performs well, the system can increase the difficulty level. If performance is poor, it can provide easier explanations and additional practice.

Slide 6: Role of PySpark

PySpark helps process data from thousands of students.

- `map()` processes individual student activities.
- `filter()` identifies weak students.
- `reduceByKey()` calculates performance scores.
- `groupByKey()` groups activities by student.

Slide 7: Benefits

- Personalized learning paths.
- Real-time feedback.
- Adaptive difficulty levels.
- Continuous improvement.

- Scalable education analytics.

5. Multi-Agent System for Disaster Management

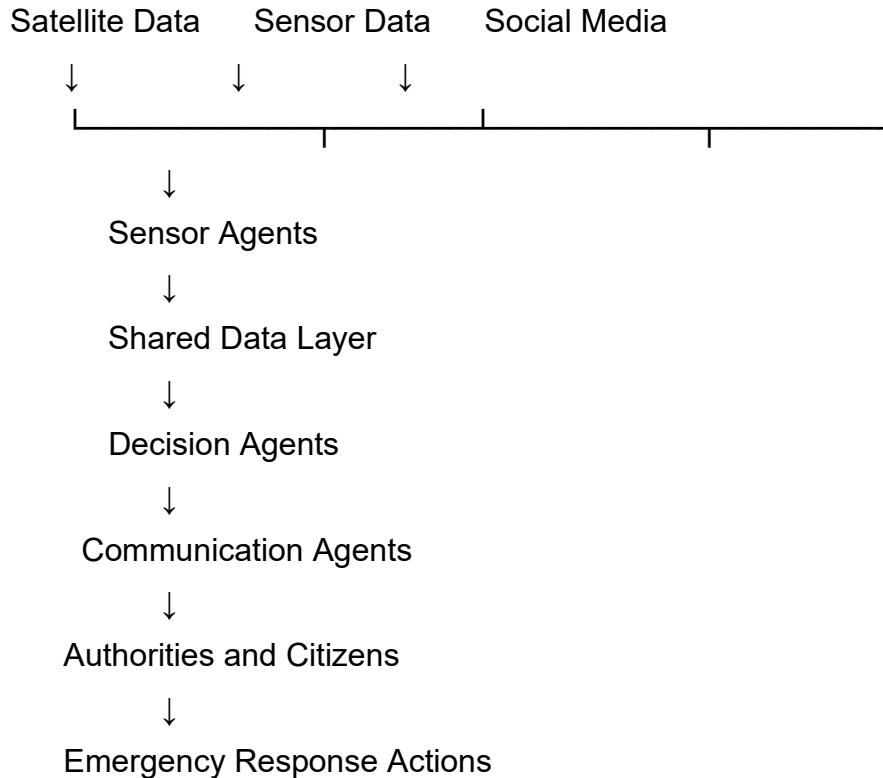
Slide 1: Title

Multi-Agent System for Smart Disaster Management

Slide 2: Introduction

A multi-agent system consists of multiple intelligent agents working together. Each agent performs a specific task and shares information with other agents.

Slide 3: System Architecture



Slide 4: Types of Agents

1. Sensor Agents

- Collect temperature, rainfall, water level, and seismic data.
- Detect unusual environmental conditions.

2. Decision Agents

- Analyze incoming information.
- Identify the type and severity of the disaster.
- Select suitable response actions.

3. Communication Agents

- Send warnings to citizens and authorities.
- Share evacuation and emergency information.

Slide 5: Agent Coordination

Sensor Agent Detects Flood Risk



Shares Data with Decision Agent



Decision Agent Confirms Risk



Communication Agent Sends Alert



Emergency Teams Respond

Slide 6: Data Sharing and Real-Time Decisions

Agents share information through a distributed data-processing system. Real-time data from different sources is combined and analyzed. This allows the system to make faster and more accurate decisions.

Slide 7: Benefits

- Early disaster detection.
- Faster emergency response.
- Better communication.
- Improved coordination.
- Reduced loss of life and property.

Conclusion

Agent Type	Application	Main Function
Reactive Agent	Traffic Control	Immediate response to traffic conditions
Goal-Based Agent	Healthcare	Achieves diagnostic and treatment goals
Utility-Based Agent	E-Commerce	Maximizes user satisfaction
Learning Agent	Education	Improves recommendations through feedback
Multi-Agent System Disaster Management		Coordinates multiple agents for response